

NYC Flights 2013 Analysis - Project Report

1. Project Overview

This project analyses the **NYC Flights 2013** dataset using Python to understand flight operations, delay patterns, airline and airport performance, and relationships between flight delays and weather conditions.

The analysis uses exploratory data analysis, data cleaning, feature engineering, statistical analysis, correlation analysis, and data visualisation to identify meaningful patterns and operational insights.

2. Business Objectives

The analysis aims to:

- Understand overall flight traffic patterns.
 - Analyse departure and arrival delays.
 - Identify differences in flight volume across months and weekdays.
 - Compare flight activity across NYC airports.
 - Evaluate airline performance based on average arrival delays.
 - Identify hourly patterns in arrival delays.
 - Examine relationships between flight characteristics and delays.
 - Explore the relationship between weather conditions and flight delays.
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3. Tools & Technologies

Programming Language: Python

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Environment: Jupyter Notebook

Key Techniques:

Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Aggregation, Statistical Analysis, Correlation Analysis, Data Visualisation

4. Data Preparation & Cleaning

The dataset was examined for missing values and data quality issues before performing the analysis.

A missing-value heatmap was used to identify patterns of incomplete data. Missing values were primarily observed in flight timing and delay-related variables such as departure time, departure delay, arrival time, arrival delay, and air time.

These missing values were addressed during preprocessing before conducting subsequent flight and delay analysis.

5. Exploratory Data Analysis

5.1 Departure Delay Distribution

The departure-delay distribution is strongly right-skewed. Most flights have little to no departure delay, while a smaller proportion experience substantially longer delays.

This indicates that extreme delays occur less frequently but can have a significant impact on overall flight performance.

5.2 Arrival Delay Distribution

Arrival delays show a similar right-skewed distribution, with most flights concentrated around low or near-zero delays.

Arrival delays are more dispersed, with a longer right tail indicating that some flights experience substantially longer delays upon arrival.

6. Flight Traffic Analysis

Flights by Month

Flight volume remains relatively consistent throughout the year.

July and August record the highest number of flights, while February has the lowest flight volume. The remaining months show relatively stable traffic levels.

Flights by Weekday

Flight volume is relatively consistent from Monday to Friday.

Saturday has the lowest number of flights, while Sunday also records lower flight volume compared with weekdays.

Flights by Origin Airport

Among the three NYC airports analysed:

- **EWR** handles the highest number of flights.
- **JFK** follows EWR.
- **LGA** records the lowest flight volume among the three.

This indicates that EWR has the highest flight traffic volume in the dataset.

7. Delay Pattern Analysis

Average Arrival Delay by Hour

Average arrival delays vary significantly by departure hour.

Delays are lowest during the late morning and early afternoon, while average delays increase steadily during the evening and reach their highest levels during late-night and early-morning hours.

This suggests that flight timing is an important factor when analysing operational delays.

Airline Performance

The top 10 carriers were compared based on average arrival delay.

F9 records the highest average arrival delay among the selected carriers, followed by **FL**, while **UA** records the lowest average arrival delay.

This indicates noticeable differences in delay performance across airlines.

8. Correlation Analysis

A correlation heatmap was used to identify relationships between numerical flight characteristics and delay-related variables.

The analysis shows strong relationships among departure and arrival timing variables and their corresponding delays.

Flight distance and air time also show a strong positive relationship, which is expected because longer flights generally require greater travel time.

Delay-related variables also show positive relationships with the derived 15-minute delay indicators.

9. Weather Analysis

Weather variables were analysed to examine their relationship with flight delays.

The analysis indicates that weather variables generally show relatively weak correlations with departure and arrival delays.

Temperature and dew point show a stronger positive relationship with each other, while visibility has a negative relationship with humidity.

Overall, the correlation analysis does not indicate a strong direct linear relationship between individual weather variables and flight delays in this dataset.

10. Key Insights

The analysis produced the following key insights:

1. **Most flights experience little or no delay**, although a smaller number experience significantly longer delays.
2. **Arrival delays show greater dispersion** and a longer right tail than typical departure delays.
3. **July and August have the highest flight volumes**, while February has the lowest.

4. **EWR has the highest flight traffic** among the three NYC airports.
 5. **Flight volume is relatively stable during weekdays**, with lower traffic on Saturday and Sunday.
 6. **Arrival delays vary by hour**, with delays increasing significantly during evening and late-night periods.
 7. **Airline delay performance varies considerably**, with F9 recording the highest average arrival delay among the selected carriers.
 8. **Flight distance and air time are strongly related**.
 9. **Weather variables show relatively weak direct correlations with flight delays** in the analysed data.
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11. Business Recommendations

Based on the analysis:

- Airlines can focus on **late-day and evening operations**, where average arrival delays increase.
 - Delay performance should be monitored **by airline and time period** to identify recurring operational issues.
 - Airports can use flight-volume patterns to improve **resource and capacity planning** during higher-traffic periods.
 - Further analysis could combine weather, airport congestion, carrier performance, and departure timing to better understand the causes of delays.
 - Predictive modelling could be explored in future work to identify flights with a higher probability of significant delays.
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12. Conclusion

This project demonstrates how Python-based exploratory data analysis can be used to understand flight operations and identify important patterns in delays, traffic volumes, airline performance, airport activity, and weather conditions.

The analysis combines data cleaning, feature engineering, statistical analysis, correlation analysis, and visualisation to transform raw flight data into meaningful operational insights.

13. Project Deliverables

File	Description
NYC_Flights_2013_Analysis.ipynb	Complete Python analysis, visualisations, and findings
NYC_Flights_2013_Project_Report.pdf	Project documentation and key business insights

